

HashMap

Key $\rightarrow$ Value

Q store population of every country?

Country name $\rightarrow$ Key (String)

population of country $\rightarrow$ value (long)

HashMap

$\rightarrow (K, V)$

$\rightarrow$ Keys has to be unique

$\rightarrow$ Key

HashSet

(K)

Key has to be unique

Key

* Frequency of Number:-

$A = \{ 2, 2, 6, 3, 8, 2, 8, 2, 3, 8, 10, 6 \}$

Q $2:3, 3:2, 4:0, 10:1$

Tc: $O(N+Q)$

Sc: $O(N)$

* first non-repeating element:-

Example

$A = \{ 4, 3, 3, 2, 5, 6, 4, 5 \} \Rightarrow 2$

Idea:-

$\rightarrow$ create a HM

$\rightarrow$ Iterate the Array and find first Key whose value = 1

Tc: $O(N)$

Sc: $O(N)$

* Distinct element in array:-

Example

$A = \{ 3, 5, 6, 5, 4 \} \rightarrow \{ 3, 5, 6, 4 \} \Rightarrow 4$

Tc: $O(N)$

Sc: $O(N)$

* Subarray sum with zero:-

$A = \{ 2, 2, 1, -3, 4, 3, 1, -2, -3, 2 \}$
 $\rightarrow$ True

Idea:-

$\rightarrow$ Find pf array $\rightarrow$ if $pf[i] = 0$ return true

$\rightarrow$ check for duplicate in the array

put everything in a HashSet

if $len(HS) < len(pf)$

there is subarray with sum 0

else

No subarray with sum 0

Tc: $O(N)$

Sc: $O(N)$